Math concepts for Data Science

DSTA

From datasets to matrices

Motivations

Activity tables show how users *map* their choices or, viceversa, how available products *map* onto their adopters.

	Matrix	Alien	Star Wars	Casablanca	Titanic
Joe	1	1	1	0	0
Jim	3	3	3	0	0
John	4	4	4	0	0
Jack	5	5	5	0	0
Jill	0	0	0	4	4
Jenny	0	0	0	5	5
Jane	0	0	0	2	2

Figure 11.6: Ratings of movies by users

Running example from Ch. 11, p. 430 of MMDS.

From tables to matrices

$$A_{7\times 5} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix}$$

$$A_{7\times 5} = \begin{pmatrix} 1 & 1 & 1 & \dots \\ 3 & 3 & 3 & \dots \\ 4 & 4 & 4 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

the matrix *'forgets'* the labels for rows/cols., eg, Joe/1st row, The Matrix/1st col. etc.

1-hot encodings

$$B_{18\times 13} = \left| \begin{array}{cccc} 1 & 0 & 0 & \ddots \\ 1 & 0 & 0 & \ddots \\ 1 & 0 & 0 & \ddots \\ & & & \ddots \end{array} \right| \begin{array}{cccc} 1 & 0 & & \ddots \\ 0 & 1 & & \ddots \\ & \ddots & & \\ & \ddots & & \end{array}$$

$$B_{18\times 13} = \left| \begin{matrix} 1 & 0 & 0 & \cdots \\ 1 & 0 & 0 & \cdots \end{matrix} \right| \begin{matrix} 1 & 0 & \cdots \\ 0 & 1 & \cdots \end{matrix} \left| \begin{matrix} 1 \\ 1 \end{matrix} \right.$$

1st col. indicates that Joe watched the film

8th col indicates that The Matrix was the film watched

the final col. is views (or ratings) from the original table: 18 reviews overall.

$$B_{18\times 13} = \left| \begin{matrix} 1 & 0 & 0 & \ddots \\ 1 & 0 & 0 & \ddots \\ 1 & 0 & 0 & \ddots \\ & & & \ddots \end{matrix} \right| \begin{matrix} 1 & 0 & \ddots \\ 0 & 1 & \ddots \\ & \ddots & \\ & \ddots & \end{matrix}$$

$U \cdot F \cdot \mathbf{r}$ (where $\cdot$ means concatenation)

Data Science as Linear Algebra

Linear equations

Q: if a user declares appreciation of Science fiction, how could it be imputed to the films they've reviewed?

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$$A\mathbf{x} = \mathbf{b}$$

use $\mathbf{r}$ instead of $\mathbf{b}$ as those are *ratings:*

$$A\mathbf{x} = \mathbf{r}$$

Interpretation: how each film contributed to determine this user's appreciation for the Sci-Fi genre

Data Science as Geometry

a matrix

- represents a linear transformation, a particular type of *mapping,* between two (linear) spaces;
- is just one of the possible representations of a mapping -it depends on a choice for the bases for source and target spaces.

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We apply linear maps (in particular, eigenvalues and eigenvectors) to matrices that *do not represent geometric transformations,* but rather some kind of relationship between entities (e.g., users and films).

datapoints are vectors

A user experience is represented by a vector: user's ratings for each film. E.g.,

$$\mathbf{joe} = \langle 1, 1, 1, 0, 0 \rangle$$

$$\mathbf{jill} = \langle 0, 0, 0, 4, 4 \rangle$$

These are *row vectors* while normally vectors are columns. The transpose T operator inverts row and columns: $\mathbf{joe}^T$ is a column vector.

$$\mathbf{joe}^T = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\mathbf{joe}^T = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

Q: Can the given users' experiences be combined to yield a specific point **r** that represents a rating of how much each user likes Sci-Fi?

$$\mathbf{r} = < 6, 9, 10, .5, 1 >^T$$

Independent vectors

Geometry sees vectors (user experiences) as axes of a reference system that *spans* a space of possible ratings.

That is possible only if at least n vectors are independent from each other.

That is automatically the case for the axes of a Cartesian diagram, or for any set of *orthogonal* vectors.

Dependent vectors

Two vectors are dependent when one is simply a multiple of the other: their direction is the same but for *stretching* or *compression*.

Dependent v. should be detected and, if possible, excluded.

Non-independence example: I only watch Jason Bourne films at my friend's

U = {Alb, Ale}, F = {The-B-id, The-B-ultimatum, …}

$$A_{Bourne} = \begin{pmatrix} 4 & 4 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}$$

The two rows are dependent! Ale depends on Alb for watching Jason Bourne films.

Test: can you find two numbers x_1 and x_2 s.t.

$$x_1 \cdot \mathbf{alb}^T + x_2 \cdot \mathbf{ale}^T = \mathbf{0}?$$

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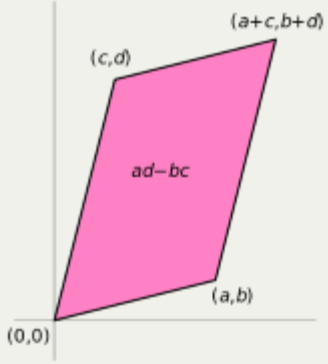
Simplest solution: $x_1 = 1$ and $x_2 = -2$.

Background: determinant

The determinant understand the matrix as an area

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$



the new space created by taking (a,b) and (c,d) as axes

if the determinant of A is zero, $|A| = 0$, then

- column vectors are not independent;
- A does not have a unique inverse matrix, so
- **it is not amenable to further processing.**

Matrix rank

The *Rank* of a matrix A is the dimension of the vector space generated by its columns.
It corresponds to

1. the maximum number of linearly-independent columns
2. the dimension of the space spanned by the rows

We consider data matrices with independent columns, ie., $rank(A_{m \times n}) = n$.

Matrix inversion, I

The identity matrix I (or U) is the unit m.: $I \cdot I = I$

$$I = \begin{pmatrix} 1 & 0 & \dots \\ 0 & 1 & \\ \vdots & & \ddots \end{pmatrix}$$

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```
import numpy as np

myI = np.eye(n)
```

creates the square identity matrix of size n.

Matrix inversion, II

Given A, find its inverse A^{-1} s.t.

$$A^{-1} \cdot A = I$$

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$$A^{-1} \cdot A = I$$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \qquad A^{-1} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}$$

A is non-singular and square: the inverse is both *left* and *right*: $A^{-1} \cdot A = A \cdot A^{-1} = I$

In Data Science, matrices are normally rectangular and inversion is a sensitive topic:

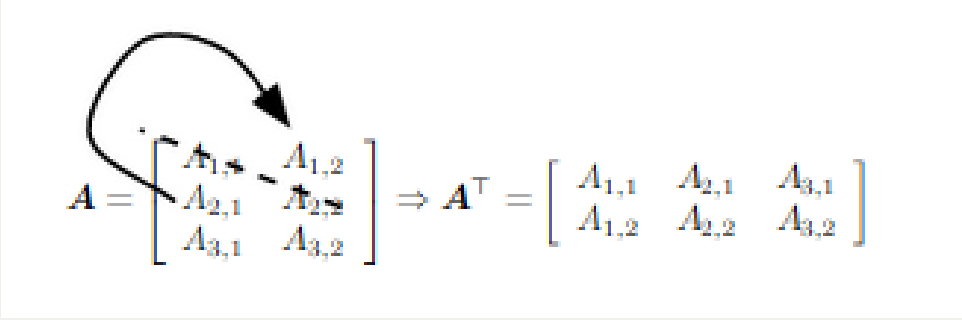
- the inverse may not exist, or be non-unique;
- numerical issues: $A^{-1} \cdot A \approx I$.

```
print(Ainverse.dot(A))

[[1.00000000e+00 0.00000000e+00]
 [1.11022302e-16 1.00000000e+00]]
```

Inversion is only defined for square matrices:

if A is rectangular then *square it:* $A' = A^T \cdot A$



transposing a rectangular m.

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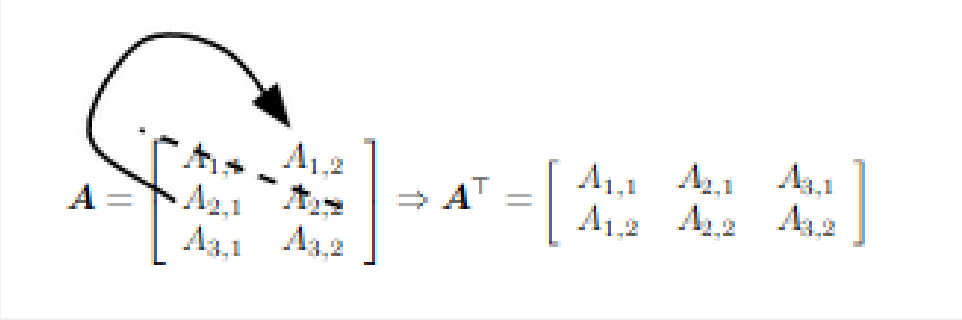


transposing a rectangular m.

```
import numpy as np

a = np.array([[11, 12], [21, 22],  [31, 32]])

at2 = a.transpose()
```

Computing

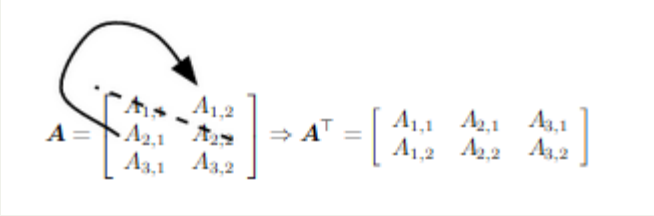
Numpy

Numpy extends Python to numerical computation.

To handle large data it creates view rather copies of arrays/matrices.

```
import numpy as np

a = np.array([[11, 12], [21, 22],  [31, 32]])

# changeable array
at2 = a.transpose()

a[0][0] = 111

print(a)

print(at2)
```

Alternative transposition:




```python
import numpy as np

a = np.array([[11, 12], [21, 22],  [31, 32]])

# tuples, not lists, and only once
at = zip(*a)

for row in at:
    print(row)
```

Matrix multiplication

```
A.dot(B) == A @ B # matrix multiplication

A.dot(B) != A * B # element-wise product
```

@ generalises to *tensors:* three-dimensional matrices.

```
m = np.array([[4, 4, 4, 0, 2], [2, 2, 2, 0, 1]])

mt = m.transpose()

mprime = m.dot(mt)

print(mprime.shape)
```

Here (the Jason Bourne ex.) rows are *not* independent.
This is revealed by $|M'| = 0$

```
print(np.linalg.det(mprime))

0.0
```

Matrix inversion and checking for errors in the results

```
i = np.eye(2)

if (np.linalg.det(mprime)):

 mprime_inv =  np.linalg.inv(mprime)

 mprime_dot_mprime_inv = mprime.dot(mprime_inv)

 # handles inf and tiny vals
 print(np.allclose(mprime_dot_mprime_inv, i))
```

prints `true` if `mprime_dot_mprime_inv` is element-wise equal to `i` within a *tolerance.*